Elk River Fish Passage

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Executive Summary

Chapter 1

Introduction

New Graph Environment was retained by the Canadian Wildlife Federation in the fall of 2020 to conduct fish passage assessments of stream crossings within the Elk River watershed upstream of the Elko Dam near Elko, BC. Hillcrest Geographics was subcontracted to provide geographical information systems (GIS) support for the project. The work was carried out in accordance with the BC Ministry of Environment: Field Assessment for Determining Fish Passage at Closed Bottom Structures, 4th Edition (BC Ministry of Environment, 2011) and Fish Passage Strategic Approach: Protocol for Prioritizing Sites for Fish Passage Remediation: 4th Edition (Fish Passage Technical Working Group, 2014).

The health and viability of freshwater fish populations can depend on access to tributary and off channel areas which provide refuge during high flows, opportunities for foraging, overwintering habitat, spawning habitat and summer rearing habitat (Bramblett et al., 2002; Swales and Levings, 1989). Culverts can present barriers to fish migration due to increased water velocity, turbulence, a vertical drop at the culvert outlet and/or maintenance issues (Slaney et al., 1997). Reconnection of fragmented habitats is a management action that can generate some of the highest ecological returns on economic investments relative to other habitat restoration techniques (Saldi-Caromile et al., 2004).

Chapter 2

Background

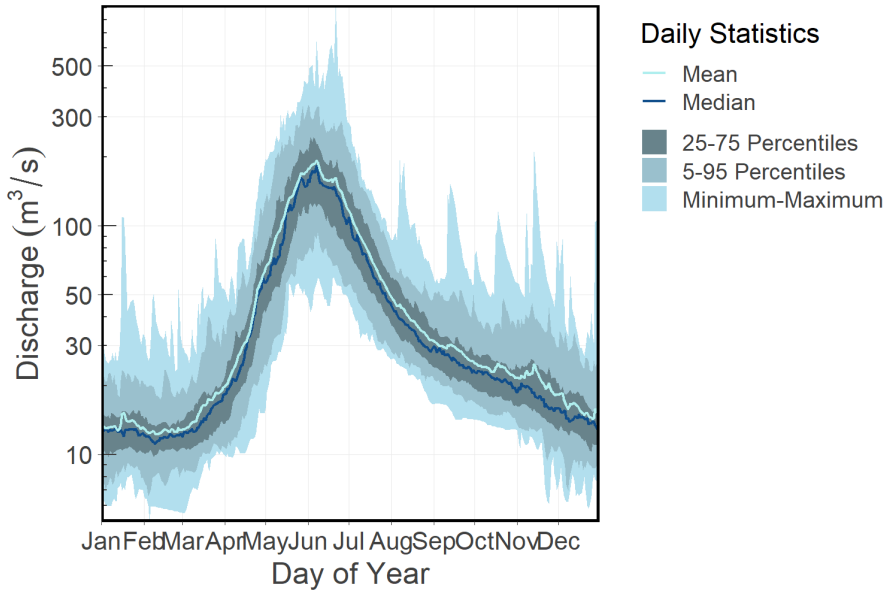
As a result of high-level direction from the provincial government, a Fish Passage Strategic Approach protocol has been developed for British Columbia to ensure that the greatest opportunities for restoration of fish passage are pursued. A Fish Passage Technical Working Group has been formed to coordinate the protocol and data is continuously amalgamated within the Provincial Steam Crossing Inventory System (PSCIS). The strategic approach protocol involves a four-phase process as described in Fish Passage Technical Working Group (2014) :

- Phase 1: Fish Passage Assessment – Fish stream crossings within watersheds with high fish values are assessed to determine barrier status of structures and document a general assessment of adjacent habitat quality and quantity.
- Phase 2: Habitat Confirmation – Assessments of crossings prioritized for follow up in Phase 1 studies are conducted to confirm quality and quantity of habitat upstream and down as well as to scope for other potential nearby barriers that could affect the practicality of remediation.
- Phase 3: Design – Site plans and designs are drawn for priority crossings where high value fish habitat has been confirmed.
- Phase 4: Remediation – Reconnection of isolated habitats through replacement, rehabilitation or removal of prioritized crossing structure barriers.

The scope of 2020/2021 project activities reported on in this document includes planning for and implementation of the first phase of fish passage assessment in the Elk River watershed upstream of the Elko Dam.

2.1 Project Location

The project was focused within the upper Elk River watershed upstream of the Elko Dam located at Elko, BC.



2.2 Fisheries

Chapter 3

Methods

3.1 Planning

We produced detailed maps of the project area identifying all locations where 1:20,000 scale TRIM map streams where intersected with roads. To determine target field sites we reviewed background habitat confirmation reports from (Masse Environmental Consultants Ltd., 2015)

which crossings on fish habitat with significant quantities of fish habitat upstream had not yet been assessed, we used the Fish Habitat Model to the estimate the quantity and quality of fish habitat potentially upstream of crossings. Using the criteria below we screen previously cross reference modelled crossing locations with sites already within the Provincial Stream Crossing Summary System. Target crossings were identified as previously unnassessed crossings on streams with likely significant quantities of fish habitat upstream.

3.2 Fish Passage Assessments

In the field, crossings surveyed included closed bottom structures (CBS), open bottom structures (OBS) and crossings considered "other" (fords, weirs, etc.). Six digit numerical crossing identifiers were generated for each crossings modeled. Crossings identified in the field that had no corresponding GIS generated ID were given unique identifiers beginning with the date in YYYYMMDD format appended with an identifier between 1 and 10 (ex. 2020091601). Photos were taken at surveyed crossings and when possible included images of the road, crossing inlet, crossing outlet, crossing barrel, channel downstream and channel upstream of the crossing and any relevant features. Additionally, the following information was recorded for all surveyed crossings: date of inspection, crossing

Table 3.1: Habitat value criteria (Fish Passage Technical Working Group, 2011).

Habitat Value	Fish Habitat Criteria
High	The presence of high value spawning or rearing habitat (e.g., locations with abundance of suitably sized gravels, deep pools, undercut banks, or stable debris) which are critical to the fish population.
Medium	Important migration corridor. Presence of suitable spawning habitat. Habitat with moderate rearing potential for the fish species present.
Low	No suitable spawning habitat, and habitat with low rearing potential (e.g., locations without deep pools, undercut banks, or stable debris, and with little or no suitably sized spawning gravels for the fish species present).

reference, crew member initials, Universal Transverse Mercator (UTM) coordinates, stream name, road name and kilometer, road tenure information, crossing type, crossing subtype, culvert diameter or span for OBS, culvert length or width for OBS. A more detailed "full assessment" was completed for all closed bottom structures.

Full assessments also included the following parameters: presence/absence of continuous culvert embedment (yes/no), average depth of embedment, whether or not the culvert bed resembled the native stream bed, presence of and percentage backwatering, fill depth, outlet drop, outlet pool depth, inlet drop, culvert slope, average downstream channel width, stream slope, presence/absence of beaver activity, presence/absence of fish at time of survey, type of valley fill, and a habitat value rating. Habitat value ratings were based on channel morphology, flow characteristics (perennial, intermittent, ephemeral), fish migration patterns, the presence/absence of deep pools, un-embedded boulders, substrate, woody debris, undercut banks, aquatic vegetation and overhanging riparian vegetation (Table 3.1). For crossings determined to be potential barriers or barriers based on the data (see section 2.3.2), a culvert fix and recommended diameter/span was proposed.

All field data collected including photos were uploaded to the Provincial Stream Crossing Inventory System (PSCIS).

3.3 Barrier Scoreing

Fish passage potential was determined for each stream crossing identified as a closed bottom structure on fish bearing and potentially fish bearing stream reaches. The combined scores from five criteria: depth and degree to which the structure is embedded, outlet drop, stream width ratio, culvert slope, and culvert length were used to screen whether each culvert was a likely barrier

Table 3.2: Fish Barrier Scoring (Fish Passage Technical Working Group 2011).

Risk	Embedded	Value	Outlet Drop (cm)	Value	SWR	Value	Slope (%)	Value	Length (m)	Value
LOW	>30cm or >20% of diameter and continuous	0	<15	0	<1.0	0	<1	0	<15	0
MOD	<30cm or 20% of diameter but continuous	5	15-30	5	1.0-1.3	3	1-3	5	15-30	3
HIGH	No embedment or discontinuous	10	>30	10	>1.3	6	>3	10	>30	6

Table 3.3: Fish Barrier Scoring Results (Fish Passage Technical Working Group 2011).

Cumlative Score	Result
0-14	passable
15-19	potential barrier
>20	barrier

to some fish species and life stages (Table 3.2, Table 3.3. These criteria were developed based on data obtained from various studies and reflect an estimation for the passage of a juvenile salmon or small resident rainbow trout (Clarkin et al., 2005, ; Bell, 1991; Thompson, 2013).

3.4 Cost Benefit Analysis

A cost benefit analysis was conducted for each crossing considered a barrier based on the amount of potential habitat to be made available by remediating fish passage at the site and an estimate of associated costs.

3.5 Habitat Gain Index

The habitat gain index is the quantity of modelled habitat upstream of the subject crossing and represents an estimate of habitat gained with remediation of fish passage at the crossing. For this project we set the threshold between fish and non-fish habitat at a gradient of 20% representing the gradient limit accessible to downstream populations. A "net" value for the index used meaning that if there is a documented PSCIS barrier crossing upstream of the subject crossing or a modelled unassessed crossing the amount of habitat is totaled to that point.

Potential options to remediate fish passage included: removal of the structure, backwatering

Cost estimates for structure replacement were generated based on the channel width, slope of the culvert, depth of fill and the road type. Base costs for installation of bridges and embedded culverts were estimated based on interviews with Phil MacDonald, Engineering Specialist FLNR - Kootenay and Steve Page,

Area Engineer - FLNR - Northern Engineering Group. Costs for installation of bridges on forest service roads was estimated at $12.5K/m and assumes that the road can be closed during construction. For streams with channel widths <2m embedded culverts can be effective with installation costs estimated at $25k

Chapter 4

Results

The analysis phase is summarized in Table 4.1

html_title <- paste0(‘### Crossing’, my_site, ’ - ‘, my_overview_info %>% pull(road_name),’ - ’, my_overview_info %>% pull(stream_name))

4.1 Results

The analysis phase is summarized in Table 4.1

\begin{table}

\caption{(#tab:cost_est)Modelled upstream habitat estimate and cost benefit.}

Table 4.1: Modelled upstream habitat estimate and cost benefit.

External ID	Stream	Road	Stream Width (m)	Priority	Fix	Cost Est ($K)	Habitat Upstream (m)	Cost Benefit (m / $K)	Cost Benefit (m2 / $K)
4600026	Tributary to Elk River	Dicken Rd	2.20	low	OBS	250	225	0.9	1.0
4600028	Bean Creek	Dicken Rd	2.00	mod	OBS	250	404	1.6	1.6
4600037	Dalzell Creek	Driveway	2.50	low	OBS	750	264	0.4	0.4
4600038	Dalzell Creek	Driveway	1.20	low	SS-CBS	150	1294	8.6	5.2
4600039	Dalzell Creek	Lower Elk Valley Road	3.80	low	OBS	750	206	0.3	0.5
4600069	Weigart Creek	Highway 43	4.30	high	OBS	2500	1206	0.5	1.0
4600070	Brule Creek	Highway 43	6.10	high	OBS	3050	763	0.3	0.8
4600080	Tributary to Elk River	Cokato Road	2.10	low	OBS	250	748	3.0	3.1
4600084	Littlemoor Creek	Lower Elk Valley Road	1.00	low	SS-CBS	150	145	1.0	0.5
4600090	Tributary to Elk River	Lower Elk Valley Road	0.00	low	SS-CBS	150	620	4.1	0.0
4600092	North Littlemoor Creek	Lower Elk Valley Road	1.50	low	SS-CBS	150	635	4.2	3.2
4600102	Tributary to Elk River	McGivern Road	0.50	low	SS-CBS	25	471	18.8	4.7
4600130	Tributary to Elk River	Cokato Road	0.65	low	SS-CBS	50	94	1.9	0.6
4600134	Tributary to	Fernie	1.40	low	SS-CBS	25	58	2.3	1.6

External ID	Stream	Road	Stream Width (m)	Priority	Fix	Cost Est ($K)	Habitat Up-stream (m)	Cost Ben-efit (m / $K)	Cost Ben-efit (m2 / $K)
4600026	Tributary to Elk River	Dicken Rd	2.20	low	OBS	250	225	0.9	1.0
4600028	Bean Creek	Dicken Rd	2.00	mod	OBS	250	404	1.6	1.6
4600037	Dalzell Creek	Driveway	2.50	low	OBS	750	264	0.4	0.4
4600038	Dalzell Creek	Driveway	1.20	low	SS-CBS	150	1294	8.6	5.2
4600039	Dalzell Creek	Lower Elk Val-ley Road	3.80	low	OBS	750	206	0.3	0.5
4600069	Weigart Creek	Highway 43	4.30	high	OBS	2500	1206	0.5	1.0
4600070	Brule Creek	Highway 43	6.10	high	OBS	3050	763	0.3	0.8
4600080	Tributary to Elk River	Cokato Road	2.10	low	OBS	250	748	3.0	3.1
4600084	Littlemoor Creek	Lower Elk Val-ley Road	1.00	low	SS-CBS	150	145	1.0	0.5
4600090	Tributary to Elk River	Lower Elk Val-ley Road	0.00	low	SS-CBS	150	620	4.1	0.0
4600092	North Lit-tle-moor Creek	Lower Elk Val-ley Road	1.50	low	SS-CBS	150	635	4.2	3.2
4600102	Tributary to Elk River	McGiverin Road	0.50	low	SS-CBS	25	471	18.8	4.7
4600130	Tributary to Elk River	Cokato Road	0.65	low	SS-CBS	50	94	1.9	0.6
4600134	Tributary to Elk	Fernie ski hill	1.40	low	SS-CBS	25	58	2.3	1.6

\end{table}

Chapter 5

Conclusion

Appendix 1 - Site Assessment Data and Photos

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.65
External ID	4600008	Length	11
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.1
Easting	640268	Resemble Channel	Yes
Northing	5481377	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Robinson Road	Fill Depth	1.2
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	1.2	Outlet Pool Depth	0
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	0.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry mostly vegetated channel. Not likely fish habitat.

Photos:
4600008![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.6
External ID	4600026	Length	22
Crew	AI, MF, DN	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	642911	Resemble Channel	No
Northing	5490630	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Dicken Rd	Fill Depth	2
Road Tenure	MoTi collector	Outlet Drop	1.3
Channel Width	2.2	Outlet Pool Depth	0.3
Stream Slope	3	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream dry at time of survey. Children's fort in upstream channel indicating that flows are likely very minimal year round.

Photos:
4600026![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.95
External ID	4600028	Length	14.5
Crew	AI, MF, DN	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	642559	Resemble Channel	No
Northing	5490377	Backwatered	No
Stream	Bean Creek	Percent Backwatered	NA
Road	Dicken Rd	Fill Depth	1.2
Road Tenure	MoTi collector	Outlet Drop	0.57
Channel Width	2	Outlet Pool Depth	0.22
Stream Slope	5	Inlet Drop	No
Beaver Activity	No	Slope	5.5
Habitat Value	High	Valley Fill	Deep Fill

Comments: Multiple fish observed in outlet pool (~190mm). Some gravels suitable for spawning located upstream.

Photos:
4600028![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.75
External ID	4600037	Length	6
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.1
Easting	651867	Resemble Channel	Yes
Northing	5522741	Backwatered	No
Stream	Dalzell Creek	Percent Backwatered	NA
Road	Driveway	Fill Depth	0.3
Road Tenure	private	Outlet Drop	0
Channel Width	2.5	Outlet Pool Depth	0
Stream Slope	1.5	Inlet Drop	No
Beaver Activity	No	Slope	2
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Embedded culvert on private driveway. EB known upstream. Culvert does not appear likely to be significantly impacting upstream fish passage for juvenile or adult WCT at time of survey.

Photos:
4600037![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.75
External ID	4600038	Length	12.5
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.18
Easting	651917	Resemble Channel	Yes
Northing	5522888	Backwatered	No
Stream	Dalzell Creek	Percent Backwatered	NA
Road	Driveway	Fill Depth	0.3
Road Tenure	private	Outlet Drop	0
Channel Width	1.2	Outlet Pool Depth	0
Stream Slope	1.5	Inlet Drop	Yes
Beaver Activity	No	Slope	2
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Culvert collapsing in the middle. Private resdience driveway. Wetland type habitat upstream. Culvert does not appear likely to be significantly impacting upstream fish passage for juvenile or adult WCT at time of survey.

Photos:
4600038![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.45
External ID	4600039	Length	16
Crew	KP, AI	Embed	Yes
UTM Zone	11	Depth Embedded	0.07
Easting	651833	Resemble Channel	Yes
Northing	5522544	Backwatered	No
Stream	Dalzell Creek	Percent Backwatered	NA
Road	Lower Elk Valley Road	Fill Depth	1
Road Tenure	MoTi arterial	Outlet Drop	0
Channel Width	3.8	Outlet Pool Depth	0.07
Stream Slope	1	Inlet Drop	No
Beaver Activity	No	Slope	3.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Three barrels (.45m each). Two of the three have water flowing through them. Upstream and downstream channel widens out, slow flow through vegetated channel. Highly manipulated banks. Deep fine substrate on both sides of crossing. Upstream resembles a wetland. Crossing does not appear likely to be significantly impacting upstream fish passage for juvenile or adult WCT at time of survey.

Photos:
4600039![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.9
External ID	4600040	Length	22
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.3
Easting	652004	Resemble Channel	Yes
Northing	5522330	Backwatered	No
Stream	Dalzell Creek	Percent Backwatered	NA
Road	Airport Road	Fill Depth	0.5
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	5	Outlet Pool Depth	0
Stream Slope	0	Inlet Drop	No
Beaver Activity	No	Slope	2.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Fenced private land upstream and downstream. Wetland area downstream. Culvert does not appear likely to be significantly impacting upstream fish passage for juvenile or adult WCT at time of survey.

Photos:
4600040![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	3.2
External ID	4600069	Length	18.2
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	650144	Resemble Channel	No
Northing	5532055	Backwatered	No
Stream	Weigart Creek	Percent Backwatered	NA
Road	Highway 43	Fill Depth	1.3
Road Tenure	MoTi highway	Outlet Drop	0.15
Channel Width	4.3	Outlet Pool Depth	0.6
Stream Slope	2	Inlet Drop	Yes
Beaver Activity	No	Slope	3.4
Habitat Value	High	Valley Fill	Deep Fill

Comments: Large stream with good flow. Flows currently mostly through south culvert which has slightly higher outlet drop. Similar elevation outlets so added together for width. Lazer level used for culvert slope. Cobble/boulder substrate. Boulder cover.

Photos:
4600069![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	2.5
External ID	4600070	Length	35
Crew	KP, AI	Embed	Yes
UTM Zone	11	Depth Embedded	0.05
Easting	651516	Resemble Channel	Yes
Northing	5528829	Backwatered	No
Stream	Brule Creek	Percent Backwatered	NA
Road	Highway 43	Fill Depth	3
Road Tenure	MoTi highway	Outlet Drop	0
Channel Width	6.1	Outlet Pool Depth	1.7
Stream Slope	1.5	Inlet Drop	No
Beaver Activity	No	Slope	2.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Dewatered upstream of highway for 600m. Deep pool present downstream fed subsurface from northside of culvert. Approximately 9 WCT in outlet pool. Four fish ~300mm and 5 fish ~200 mm. High confidence that the larger fish were WCT as easy to recognize spotted backs from bank.

Photos:
4600070![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.75
External ID	4600077	Length	20
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.1
Easting	639864	Resemble Channel	Yes
Northing	5483627	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Mt Mclean Road	Fill Depth	4
Road Tenure	Fernie local	Outlet Drop	0
Channel Width	2.7	Outlet Pool Depth	0
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	0.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry channel. Culvert is embedded.

Photos:
4600077![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.1
External ID	4600080	Length	43
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	640568	Resemble Channel	No
Northing	5481516	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Cokato Road	Fill Depth	2
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	2.1	Outlet Pool Depth	0
Stream Slope	3.5	Inlet Drop	No
Beaver Activity	No	Slope	4
Habitat Value	Low	Valley Fill	Deep Fill

Comments: 2 barrels, 0.90 and 1.1m. Debris rack on inlet. Stream dry at time of survey.

Photos:
4600080![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	4600084	Length	22
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651532	Resemble Channel	No
Northing	5521052	Backwatered	No
Stream	Littlemoor Creek	Percent Backwatered	NA
Road	Lower Elk Valley Road	Fill Depth	0.8
Road Tenure	MoTi arterial	Outlet Drop	0.2
Channel Width	1	Outlet Pool Depth	0.15
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	3
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Flows through fenced private land upstream and downstream. Inlet has quite a bit of debris on it. Upstream is much steeper than downstream with gradient estimated at 12%. Small stream with good flow.

Photos:
4600084![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.45
External ID	4600090	Length	18
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651814	Resemble Channel	No
Northing	5519652	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Lower Elk Valley Road	Fill Depth	1
Road Tenure	MoTi arterial	Outlet Drop	0
Channel Width	0	Outlet Pool Depth	0
Stream Slope	0	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: No visible channel. Not likely fish habitat.
Agricultural area.
Photos:
4600090![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culv
PSCIS ID	NA	Diameter	0.8
External ID	4600092	Length	17.5
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651701	Resemble Channel	No
Northing	5521881	Backwatered	No
Stream	North Littlemoor Creek	Percent Backwatered	NA
Road	Lower Elk Valley Road	Fill Depth	0.6
Road Tenure	MoTi arterial	Outlet Drop	0
Channel Width	1.5	Outlet Pool Depth	0
Stream Slope	2.5	Inlet Drop	No
Beaver Activity	No	Slope	3
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream slope estimated due to fenced private land.
Photos:
4600092![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.5
External ID	4600102	Length	16
Crew	AI, Kp	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	640306	Resemble Channel	No
Northing	5481672	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	McGiverin Road	Fill Depth	1
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	0.5	Outlet Pool Depth	0
Stream Slope	0	Inlet Drop	No
Beaver Activity	No	Slope	1
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Vegetated channel. Not likely fish habitat.
Photos:
4600102![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.2
External ID	4600130	Length	22
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	640037	Resemble Channel	No
Northing	5483655	Backwatered	Yes
Stream	Tributary to Elk River	Percent Backwatered	20
Road	Cokato Road	Fill Depth	8
Road Tenure	Fernie collector	Outlet Drop	0
Channel Width	0.65	Outlet Pool Depth	0
Stream Slope	9	Inlet Drop	Yes
Beaver Activity	No	Slope	9
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Very low flow at time of survey and 0.5m inlet drop
from debris. Very steep culvert, possibly 2 sections with further
upstream section being much steeper.
Photos:
4600130![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.9
External ID	4600134	Length	48
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	638639	Resemble Channel	No
Northing	5480681	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Fernie ski hill	Fill Depth	3
Road Tenure	unclassified	Outlet Drop	0.09
Channel Width	1.4	Outlet Pool Depth	0.22
Stream Slope	5	Inlet Drop	No
Beaver Activity	No	Slope	9
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Debris guard on inlet. Unable to see through the culvert to other end. Highly manipulated banks, drains large parking area and ski hill. Steep with placed rocks (rip rap) in spots upstream. Stream goes under large turn around, paved area. Culvert likely >1 piece with slight angle. Armoured banks downstream, lots of sediment (sand) form pavement and parking area.

Photos:
4600134![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.9
External ID	4600140	Length	22
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651110	Resemble Channel	No
Northing	5515356	Backwatered	No
Stream	Whiting Creek	Percent Backwatered	NA
Road	Highway 43	Fill Depth	3
Road Tenure	MoTi arterial	Outlet Drop	0.18
Channel Width	0.6	Outlet Pool Depth	0
Stream Slope	2	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: No access up or downstream due to fenced private land.
Stream slope estimated. Water pipe intake at the outlet.
Photos:
4600140![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Pipe Arch
PSCIS ID	NA	Diameter	4
External ID	4600157	Length	44
Crew	AI, MF, DN	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	643565	Resemble Channel	NA
Northing	5490325	Backwatered	NA
Stream	Hartley Creek	Percent Backwatered	NA
Road	Highway 3	Fill Depth	NA
Road Tenure	MoTi highway	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: This crossing is very full of debris and has been
recently dredges. Less than 30cm freeboard from top of aggraded
gravels at inlet and top of pipe arch. Newly installed overflow
pipe (0.55 diameter) on left bank.
Photos:
4600157![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.8
External ID	4600158	Length	29.5
Crew	AI, MF, DN	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	642739	Resemble Channel	No
Northing	5489444	Backwatered	No
Stream	Bean Creek	Percent Backwatered	NA
Road	Highway 3	Fill Depth	1.5
Road Tenure	MoTi highway	Outlet Drop	0.17
Channel Width	3.2	Outlet Pool Depth	0.24
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	1
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Fish observed at upstream crossing on Bean Rd. Grate on inlet may prevent passage off adult fish. Fill depth estimated from photos.

Photos:
4600158![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1
External ID	4600169	Length	36
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	638850	Resemble Channel	No
Northing	5480833	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Highline Drive (Fernie ski hill)	Fill Depth	6
Road Tenure	MoTi local	Outlet Drop	0.2
Channel Width	2.3	Outlet Pool Depth	0.25
Stream Slope	11.5	Inlet Drop	No
Beaver Activity	No	Slope	13
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Very recent heavy rains, 2 side channels for ski hill drainage infrastructure giving significant flow contributions just upstream of crossing, one of which is especially turbid (from large parking lot). Culvert not baffled, very steep. A lot of fill on paved access to condos. Old metal collar of culvert ripped off and in outlet pool.

Photos:
4600169![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.6
External ID	4600181	Length	73
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	652322	Resemble Channel	No
Northing	5527529	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Line creek mine road	Fill Depth	3.5
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	0.5	Outlet Pool Depth	0
Stream Slope	2	Inlet Drop	No
Beaver Activity	No	Slope	2
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry at time of survey, no channel present, area vegetated grassland. Two culverts, 0.60m in diameter. Top culvert 0.5m higher than the lower.

Photos:
4600181![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Oval Culvert
PSCIS ID	NA	Diameter	3.3
External ID	4600183	Length	20
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651626	Resemble Channel	No
Northing	5528888	Backwatered	No
Stream	Brule Creek	Percent Backwatered	NA
Road	Busato Road	Fill Depth	1
Road Tenure	MoTi local	Outlet Drop	0.7
Channel Width	7.1	Outlet Pool Depth	1.5
Stream Slope	2	Inlet Drop	Yes
Beaver Activity	No	Slope	4
Habitat Value	High	Valley Fill	Deep Fill

Comments: Large stream with good flow at this crossing. Culvert inlet is damaged and there is a large outlet drop. Upstream of highway the stream is subsurface for ~600m.

Photos:
4600183![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	4600184	Length	30
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	650954	Resemble Channel	No
Northing	5522199	Backwatered	No
Stream	North Littlemoor Creek	Percent Backwatered	NA
Road	Highway 43	Fill Depth	4
Road Tenure	MoTi highway	Outlet Drop	0.65
Channel Width	1.6	Outlet Pool Depth	0.55
Stream Slope	5	Inlet Drop	No
Beaver Activity	No	Slope	8
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Small stream with good flow, EB known upstream. Fenced private land upstream and downstream. Stream measurements estimated due to fenced private land. Large outlet drop.

Photos:
4600184![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.9
External ID	4600185	Length	40
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651002	Resemble Channel	No
Northing	5521022	Backwatered	No
Stream	Littlemoor Creek	Percent Backwatered	NA
Road	Highway 43	Fill Depth	5
Road Tenure	MoTi highway	Outlet Drop	0.3
Channel Width	1.2	Outlet Pool Depth	0.55
Stream Slope	3.5	Inlet Drop	Yes
Beaver Activity	No	Slope	5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Culvert has extension with corner, small stream with good flow. WCT and EB documented upstream. Gravels present suitable for WCT and EB spawning.

Photos:
4600185![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.8
External ID	4600186	Length	22
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	651051	Resemble Channel	No
Northing	5519343	Backwatered	No
Stream	Hollow Creek	Percent Backwatered	NA
Road	Highway 43	Fill Depth	0.8
Road Tenure	MoTi highway	Outlet Drop	0.46
Channel Width	1.1	Outlet Pool Depth	0.36
Stream Slope	2	Inlet Drop	No
Beaver Activity	No	Slope	4
Habitat Value	Low	Valley Fill	Deep Fill

Comments: No access up or downstream due to fenced private land.
Stream slope estimated.
Photos:
4600186![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	1.2
External ID	4600316	Length	12
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	641167	Resemble Channel	No
Northing	5479429	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Cokato Road	Fill Depth	0.5
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	4.1	Outlet Pool Depth	0
Stream Slope	4	Inlet Drop	No
Beaver Activity	No	Slope	2.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream dry at time of survey. Three barrels, 1.2m in
diameter. All three very similar elevation, all perched .2-.50m
above substrate.
Photos:
4600316![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-17	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	4600329	Length	12
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	652325	Resemble Channel	Yes
Northing	5515789	Backwatered	No
Stream	Tributary to Whiting Creek	Percent Backwatered	NA
Road	Lower Elk Valley Road	Fill Depth	1
Road Tenure	MoTi arterial	Outlet Drop	0
Channel Width	0.5	Outlet Pool Depth	0
Stream Slope	0	Inlet Drop	No
Beaver Activity	No	Slope	1
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Whiting creek appeared to not be present through agricultural field downstream. This appeared to be only channel present for Whiting crossing this road although the main Whiting channel is present upstream. Dry, heavily vegetated channel downstream. Defined channel upstream but seems unlikely to be fish bearing.

Photos:
4600329![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	1.2
External ID	4600332	Length	25
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	639511	Resemble Channel	No
Northing	5481114	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Highway 3	Fill Depth	2.3
Road Tenure	MoTi highway	Outlet Drop	0
Channel Width	3.3	Outlet Pool Depth	0.5
Stream Slope	4	Inlet Drop	Yes
Beaver Activity	No	Slope	1.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Fry observed at outlet pool. Inlet of active pipe mostly plugged with debris. Overflow pipe is 0.8m outlet drop with fry/juveniles stranded in outlet pool. Flows out of Fernie Ski Hill Road development area.

Photos:
4600332![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Oval Culvert
PSCIS ID	NA	Diameter	2.6
External ID	4600367	Length	20
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	643534	Resemble Channel	No
Northing	5490723	Backwatered	No
Stream	Hartley Creek	Percent Backwatered	NA
Road	Dicken Road	Fill Depth	0.4
Road Tenure	MoTi collector	Outlet Drop	0.4
Channel Width	3.5	Outlet Pool Depth	0.8
Stream Slope	1	Inlet Drop	No
Beaver Activity	No	Slope	2
Habitat Value	High	Valley Fill	Deep Fill

Comments: Laser level used for culvert slope. WCT spawning system. See Management Plan for the Westslope Cutthroat Trout (Oncorhynchus clarkii lewisi) in British Columbia 2014 http://a100.gov.bc.ca/pub/eirs/finishDownloadDocument.do?subdocumentId=9781

Photos: 4600367![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-20	Crossing Sub Type	Ford
PSCIS ID	NA	Diameter	NA
External ID	4601129	Length	NA
Crew	KP, AI	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	661062	Resemble Channel	NA
Northing	5524446	Backwatered	NA
Stream	Grave Creek	Percent Backwatered	NA
Road	NA	Fill Depth	NA
Road Tenure	unclassified	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Deactivated road. Ford.

Photos: 4601129![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Ford
PSCIS ID	NA	Diameter	NA
External ID	4601205	Length	NA
Crew	AI, KP	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	650917	Resemble Channel	NA
Northing	5528513	Backwatered	NA
Stream	Brule Creek	Percent Backwatered	NA
Road	Private	Fill Depth	NA
Road Tenure	private	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Pulled crossing, no issues, is now a ford.
Photos:
4601205![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.9
External ID	4601556	Length	14.5
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	638522	Resemble Channel	No
Northing	5480616	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Fernie ski hill	Fill Depth	1.2
Road Tenure	unclassified	Outlet Drop	0
Channel Width	1.3	Outlet Pool Depth	0.2
Stream Slope	8	Inlet Drop	Yes
Beaver Activity	No	Slope	9
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream gradient immediately upstream of crossing is near 20%. Crossing is last of a series within close proximity to each other. The crossing downstream of this one is near 100m long. Likely more crossings upstream on ski hill infrastructure and access roads. Recent heavy rains.
Photos:
4601556![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.5
External ID	4601594	Length	22
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	641090	Resemble Channel	No
Northing	5479392	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Railway	Fill Depth	2
Road Tenure	Canadian Pacific	Outlet Drop	0
Channel Width	2.7	Outlet Pool Depth	0
Stream Slope	2.5	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream dry at time of survey. 2 barrels (0.80 and 1.5m in diameter), similar heights but 1.5m slightly lower. Smaller barrel inlet damaged.

Photos:
4601594![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.2
External ID	4601639	Length	99
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	638630	Resemble Channel	No
Northing	5480655	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Fernie ski hill	Fill Depth	2.5
Road Tenure	unclassified	Outlet Drop	0.73
Channel Width	1.5	Outlet Pool Depth	0.65
Stream Slope	6	Inlet Drop	Yes
Beaver Activity	No	Slope	11
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Huge crossing through the ski hill parking lot and lodge area. Largeoutlet drop and pool, extremely long culvert (close to 100m) tied in with other drainage infrastructure at the base of the Fernie ski hill. Inlet drop about .15m.

Photos:
4601639![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-20	Crossing Sub Type	Round Cu
PSCIS ID	NA	Diameter	0.9
External ID	4602270	Length	14
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	655441	Resemble Channel	No
Northing	5524175	Backwatered	No
Stream	Tributary to Grave Creek	Percent Backwatered	NA
Road	NA	Fill Depth	2
Road Tenure	unclassified	Outlet Drop	0
Channel Width	1.5	Outlet Pool Depth	0
Stream Slope	3	Inlet Drop	Yes
Beaver Activity	No	Slope	4
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Inlet clogged with debris 3/4 of way up. Upstream area is wetland/pond.

Photos: 4602270![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Ford
PSCIS ID	NA	Diameter	NA
External ID	4602276	Length	NA
Crew	AI	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	649758	Resemble Channel	NA
Northing	5527935	Backwatered	NA
Stream	Brule Creek	Percent Backwatered	NA
Road	Spur	Fill Depth	NA
Road Tenure	unclassified	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Deactivated. Ford. no issues.

Photos: 4602276![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.2
External ID	4602349	Length	10.5
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	638525	Resemble Channel	No
Northing	5481496	Backwatered	Yes
Stream	Tributary to Elk River	Percent Backwatered	20
Road	Fernie Nordic Trail	Fill Depth	1.5
Road Tenure	unclassified	Outlet Drop	0
Channel Width	2	Outlet Pool Depth	2
Stream Slope	7	Inlet Drop	No
Beaver Activity	No	Slope	3.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Outlet pool appears dredged, very deep, approximately 2m and 7-8m long. Nice stream, good flow.

Photos:

4602349![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-20	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	4602533	Length	7.5
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	661172	Resemble Channel	No
Northing	5524451	Backwatered	No
Stream	Grave Creek	Percent Backwatered	NA
Road	NA	Fill Depth	0.35
Road Tenure	unclassified	Outlet Drop	0
Channel Width	0.1	Outlet Pool Depth	0
Stream Slope	1	Inlet Drop	No
Beaver Activity	No	Slope	4
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream dry at time of survey, channel not visible for about 40 downstream. Water and channel appear about 65m downstream of crossing. Upstream side of crossing inlet not visible, covered by road fill or deep organic debris. Channel upstream non-existent and dry. This system (labelled as Grave creek on the map) must be a side channel or remnant channel.

Photos:
4602533![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.6
External ID	4603265	Length	13
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	640287	Resemble Channel	No
Northing	5481650	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Railway	Fill Depth	3
Road Tenure	Canadian Pacific	Outlet Drop	0
Channel Width	0.5	Outlet Pool Depth	0
Stream Slope	0	Inlet Drop	No
Beaver Activity	No	Slope	0.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry vegetated channel. Not likely fish habitat.

Photos:
4603265![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.2
External ID	4603291	Length	13
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	640891	Resemble Channel	No
Northing	5480517	Backwatered	No
Stream	Cokato Creek	Percent Backwatered	NA
Road	Cokato Road	Fill Depth	0.5
Road Tenure	MoTi local	Outlet Drop	0
Channel Width	4.5	Outlet Pool Depth	0
Stream Slope	4	Inlet Drop	No
Beaver Activity	No	Slope	4
Habitat Value	Low	Valley Fill	Deep Fill

Comments: 3 large barrels (all 1.2m), very perched 0.7-1.3m above substrate. Stream dry at time of survey. Debris rack on upstream side of culvert. Note there is a slope change at the inlet of structure, much steeper short section possibly from damage.

Photos:
4603291![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.8
External ID	4604198	Length	9
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	647819	Resemble Channel	No
Northing	5498551	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Hadner FSR	Fill Depth	1
Road Tenure	FLNR 6946	Outlet Drop	0.6
Channel Width	2.9	Outlet Pool Depth	0.3
Stream Slope	19	Inlet Drop	Yes
Beaver Activity	No	Slope	6
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Steep stream with good flow. FISS sample site 2593 near crossing location. Suspect too steep to be fish bearing.

Photos:
4604198![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-24	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.9
External ID	4604455	Length	16
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	638669	Resemble Channel	No
Northing	5480601	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Fernie ski hill	Fill Depth	2
Road Tenure	unclassified	Outlet Drop	0.15
Channel Width	1.5	Outlet Pool Depth	0.2
Stream Slope	9	Inlet Drop	Yes
Beaver Activity	No	Slope	7
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Inlet drop about .40m, significant and looks like
increased barrier. Crossing is Fernie ski hill gravel road.
urrently high and turbid due to heavy rain.
Photos:
4604455![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Pipe Arch
PSCIS ID	NA	Diameter	2.5
External ID	4605636	Length	14.5
Crew	KP, AI	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	644148	Resemble Channel	NA
Northing	5564425	Backwatered	NA
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	NA
Road Tenure	FLNR 0103	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Appears passable.
Photos:
4605636![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.9
External ID	4605649	Length	17
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	645873	Resemble Channel	No
Northing	5556758	Backwatered	Yes
Stream	Tributary to Elk River	Percent Backwatered	100
Road	Elk River FSR	Fill Depth	1.5
Road Tenure	FLNR 0103	Outlet Drop	0
Channel Width	1	Outlet Pool Depth	0.6
Stream Slope	2.5	Inlet Drop	No
Beaver Activity	No	Slope	0.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Low, slow flow. Culvert completely backwatered.

Photos:

4605649![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.9
External ID	4605653	Length	10.92
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	644666	Resemble Channel	No
Northing	5564940	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	0.4
Road Tenure	FLNR 0103	Outlet Drop	0
Channel Width	2.5	Outlet Pool Depth	0.2
Stream Slope	2	Inlet Drop	No
Beaver Activity	No	Slope	3.2
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Good habitat, habitat assessment (FHAP) conducted in 2016 (Masse Environmental Consultants Ltd.) for proposed coal mine EA (Bingay). Electrofished U/S and D/S, see provincial records and data submission file at http://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=52717 . Culvert does not appear to be barrier to any species at any life stage at time of survey. Could be barrier to small fish at high flows. EB and WCT recrded upstream.

Photos:
4605653![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	4605675	Length	10.5
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	645225	Resemble Channel	No
Northing	5567096	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	0.5
Road Tenure	NA	Outlet Drop	0
Channel Width	2	Outlet Pool Depth	0.05
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	2.5
Habitat Value	Medium	Valley Fill	Deep Fill

Comments: Culvert slope estimate. Stream dredged upstream to remove sediments. Flows are slow through culvert and unlikely barrier for fry/parr for much of year. Crossing 4606244 is on deactivated spur upstream so very likely a ford, about 300m upstream same thing for 4604099.

Photos:
4605675![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.5
External ID	4605697	Length	1.8
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	648722	Resemble Channel	No
Northing	5548198	Backwatered	No
Stream	Crossing Creek	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	1
Road Tenure	FLNR 0103	Outlet Drop	0
Channel Width	2.5	Outlet Pool Depth	0
Stream Slope	3.5	Inlet Drop	No
Beaver Activity	No	Slope	5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry channel at time of survey, channel transitions to grass flats about 60 meters downstream.

Photos:
4605697![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.35
External ID	4605705	Length	15
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	643733	Resemble Channel	No
Northing	5560586	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	1.2
Road Tenure	Unknown	Outlet Drop	2.15
Channel Width	1	Outlet Pool Depth	0.4
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	2.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream not located where mapped. Unlikely fish bearing.
Flows into forest floor downstream of culvert.
Photos:
4605705![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.6
External ID	4605707	Length	11
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	643981	Resemble Channel	No
Northing	5561132	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	0.4
Road Tenure	FLNR 0103	Outlet Drop	0
Channel Width	2.3	Outlet Pool Depth	0
Stream Slope	1	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Stream dry at time of survey. Vegetated channel, loaded
with sediment. Inlet of culvert damaged, partly bent. Inlet and
outlet area dredged. Not likely fish habitat.
Photos:
4605707![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	4605708	Length	10
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.1
Easting	643611	Resemble Channel	Yes
Northing	5559835	Backwatered	No
Stream	Tributary to Lowe Creek	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	0.5
Road Tenure	FLNR 0103	Outlet Drop	0
Channel Width	1.1	Outlet Pool Depth	0
Stream Slope	3.5	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Channelized downstream due to dredging for about 30m.
Small stream, substrate primarily fines but some gravel present.
Photos:
4605708![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.5
External ID	4605732	Length	49
Crew	KP, AI	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	646735	Resemble Channel	No
Northing	5554534	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	8
Road Tenure	FLNR 0103	Outlet Drop	1.48
Channel Width	3.5	Outlet Pool Depth	1.3
Stream Slope	1.5	Inlet Drop	Yes
Beaver Activity	No	Slope	3.5
Habitat Value	High	Valley Fill	Deep Fill

Comments: Large perched culvert. High value habitat. Habitat
confirmation and sampling conducted.
Photos:
4605732![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	1.2
External ID	4605733	Length	9.5
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	647191	Resemble Channel	No
Northing	5552693	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	1
Road Tenure	FLNR 0103	Outlet Drop	0.3
Channel Width	3.1	Outlet Pool Depth	0.6
Stream Slope	2	Inlet Drop	No
Beaver Activity	No	Slope	1
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Freshly dredged. Dry, Dredging may extend well down channel. See photo from about 130m downstream.

Photos:
4605733![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	1.5
External ID	4605742	Length	15
Crew	KP, AI	Embed	Yes
UTM Zone	11	Depth Embedded	0.12
Easting	643578	Resemble Channel	Yes
Northing	5560087	Backwatered	No
Stream	Lowe Creek	Percent Backwatered	NA
Road	Elk River FSR	Fill Depth	0.3
Road Tenure	FLNR 0103	Outlet Drop	0
Channel Width	2.5	Outlet Pool Depth	0.2
Stream Slope	3.5	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	High	Valley Fill	Deep Fill

Comments: Upstream past dredging natural channel 2.5m wide. Downstream below dredging extremely braided, original channel difficult to track. Channel widths >10m at times with substrate on top of banks and in bases of trees. Channel dredged upstream and downstream approximately 75m on each. Habitat assessment (FHAP) conducted in 2016 (Masse Environmental Consultants Ltd.) for proposed coal mine EA (Bingay). Electrofished D/S, see provincial records and data submission file at http://a100.gov.bc.ca/pub/acat/public/viewReport.do?reportId=52717 .

Photos: 4605742![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Cu
PSCIS ID	NA	Diameter	0.4
External ID	4606669	Length	10
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	646462	Resemble Channel	No
Northing	5554360	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Spur from Elk River FSR	Fill Depth	1
Road Tenure	Canfor R08473	Outlet Drop	0
Channel Width	1	Outlet Pool Depth	0
Stream Slope	0.5	Inlet Drop	No
Beaver Activity	No	Slope	2
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Plan in place to deactivate entire network here (personal communication with local machinery operator). Dry, vegetated channel. Likely non-fish bearing.

Photos: 4606669![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Concrete Box
PSCIS ID	NA	Diameter	1.2
External ID	4606807	Length	9
Crew	KP, AI	Embed	Yes
UTM Zone	11	Depth Embedded	0.09
Easting	639952	Resemble Channel	No
Northing	5483636	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Railway	Fill Depth	6
Road Tenure	Canadian Pacific	Outlet Drop	0
Channel Width	1.5	Outlet Pool Depth	0
Stream Slope	30	Inlet Drop	No
Beaver Activity	No	Slope	1.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Concrete arch, >30% downstream of culvert for 10m likely from railway fill. Dry downstream with stagnant pools upstream.

Photos: 4606807![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-16	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.4
External ID	4606835	Length	10
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	644829	Resemble Channel	No
Northing	5559116	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Spur from Elk River FSR	Fill Depth	1
Road Tenure	Canfor R08370	Outlet Drop	0.3
Channel Width	1	Outlet Pool Depth	0
Stream Slope	3	Inlet Drop	No
Beaver Activity	No	Slope	2.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Road is deactivated downstream so no crossings below (4607172, 4606929). Dry vegetated channel. Not likely fish bearing at this location.

Photos:
4606835![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Bridge
PSCIS ID	NA	Diameter	4
External ID	4607023	Length	18
Crew	AI, KP	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	648041	Resemble Channel	NA
Northing	5499983	Backwatered	NA
Stream	McCool Creek	Percent Backwatered	NA
Road	NA	Fill Depth	NA
Road Tenure	Canfor R08477	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Bridge.

Photos:
4607023![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master/da

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-18	Crossing Sub Type	Bridge
PSCIS ID	NA	Diameter	4
External ID	2020091801	Length	18
Crew	AI, KP	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	648381	Resemble Channel	NA
Northing	5499536	Backwatered	NA
Stream	McCool Creek	Percent Backwatered	NA
Road	Hadner FSR	Fill Depth	NA
Road Tenure	FLNR 6946	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Bridge.
Photos:
2020091801![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-22	Crossing Sub Type	Bridge
PSCIS ID	NA	Diameter	1.2
External ID	2020092201	Length	12
Crew	AI, KP	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	634969	Resemble Channel	NA
Northing	5484828	Backwatered	NA
Stream	Tributary to Lizard Creek	Percent Backwatered	NA
Road	Lazy Lizard Lower	Fill Depth	NA
Road Tenure	Unknown	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Bike trail.
Photos:
2020092201![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Bridge
PSCIS ID	NA	Diameter	1.2
External ID	2020092301	Length	8
Crew	AI, KP	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	636474	Resemble Channel	NA
Northing	5483873	Backwatered	NA
Stream	Tributary to Lizard Creek	Percent Backwatered	NA
Road	Trail	Fill Depth	NA
Road Tenure	Unknown	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Trail. Stream has been diverted to beside the road.
Photos:
2020092301![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culvert
PSCIS ID	NA	Diameter	0.6
External ID	2020092302	Length	14
Crew	AI, KP	Embed	Yes
UTM Zone	11	Depth Embedded	0.03
Easting	640218	Resemble Channel	Yes
Northing	5481065	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Hill Road	Fill Depth	1.2
Road Tenure	Unknown	Outlet Drop	0
Channel Width	1.5	Outlet Pool Depth	0
Stream Slope	0	Inlet Drop	No
Beaver Activity	No	Slope	0.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry vegetated channel. Not likely fish habitat.
Photos:
2020092302![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/raw/master

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Round Culve
PSCIS ID	NA	Diameter	0.6
External ID	2020092303	Length	15
Crew	AI, KP	Embed	No
UTM Zone	11	Depth Embedded	NA
Easting	640227	Resemble Channel	No
Northing	5481028	Backwatered	No
Stream	Tributary to Elk River	Percent Backwatered	NA
Road	Driveway	Fill Depth	1
Road Tenure	Unknown	Outlet Drop	0
Channel Width	1.5	Outlet Pool Depth	0
Stream Slope	2	Inlet Drop	No
Beaver Activity	No	Slope	0.5
Habitat Value	Low	Valley Fill	Deep Fill

Comments: Dry, mostly vegetated channel. Not likely fish habitat.
Two pipes, one buried by fill. Inlet elevated resulting in small
are of wetland vegetation upstream.
Photos:
2020092303![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/

Location and Stream Data	-	Crossing Characteristics	–
Date	2020-09-23	Crossing Sub Type	Bridge
PSCIS ID	NA	Diameter	1.2
External ID	2020092310	Length	13.5
Crew	KP, AI	Embed	NA
UTM Zone	11	Depth Embedded	NA
Easting	636029	Resemble Channel	NA
Northing	5484419	Backwatered	NA
Stream	Tributary to Lizard Creek	Percent Backwatered	NA
Road	Lazy Lizard Lower	Fill Depth	NA
Road Tenure	Unknown	Outlet Drop	NA
Channel Width	NA	Outlet Pool Depth	NA
Stream Slope	NA	Inlet Drop	NA
Beaver Activity	No	Slope	NA
Habitat Value	NA	Valley Fill	NA

Comments: Nice wooden pedestrian/bike bridge for recreational use.
Photos:
2020092310![](https://github.com/NewGraphEnvironment/fish_passage_elk_2020_reporting/

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